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SUMMARY

At Corient, a global wealth manager with about $508B in client assets, I design and build hybrid, local-first AI for confidential client data: on-premises LLM architecture, GPU strategy, Azure data residency and Power Platform automation. I trained as an aerospace engineer (Master's in Space Engineering, Michigan, GPA 4.01; B.S. in Mechanical Engineering, Boston University, cum laude).

EXPERIENCE

Corient | Neuchâtel, Switzerland — Jun 2026 – Present

Associate, AI and Automation Analyst, Business Systems (IT Integrations)

- Wrote the 80-page architecture and implementation plan for Corient's hybrid AI platform: client data stays on-premises behind an Envoy gateway with a fail-closed policy service, vLLM serves open models, the local LiteLLM pool has no internet route, Temporal runs agent workflows with deterministic outcomes, and pgvector with row-level security holds retrieval data. Included a five-year cost comparison of five delivery options.
- Wrote decision papers for technology leadership making the case for local AI as a confidentiality and resilience failsafe: sized outage exposure from 266 public LLM-API incidents over 12 months, and costed OCR of a 30M+ document archive across Azure Document Intelligence, Copilot Credits and local vision models.
- Compared eight on-premises options, from a $4K desktop AI unit to a $340K GPU rack, and set a staged roadmap that opens with a reversible $17K pilot below the capex threshold; proposed a five-unit dev/test fleet (NVIDIA DGX Spark, AMD Strix Halo) to prove workloads locally before Azure.
- Mapped Azure AI hosting for clients under Jersey, UK, Swiss, South African and US law: compared serverless, batch, provisioned-throughput and dedicated-GPU deployments on cost, and set residency rules (region-pinned deployments, no global routing of client data, enforced by Azure Policy). Specified a self-hosted coding assistant on Azure (A100, Ollama, Qwen2.5-Coder) for 30–50 developers.
- Scored 50 local-LLM use cases in a 53-page portfolio and selected a ten-item first wave. Wrote a 61-page Power Automate Flow Catalogue with 12 reusable patterns and 8 build-ready flows for the COO's office.
- Delivered Outlook email and calendar auto-categorisation for the COO's office (Microsoft Graph, 62 rules, 15-minute sweep, rules held in SharePoint lists); built a Laserfiche and Power Automate pipeline that splits batch invoices into single pages and extracts 13 payment fields from each, and a PDF signature-stamping engine (Python, PyMuPDF, packaged executable, Laserfiche Workflow C# script) tested on a 40-document corpus.
- Built voice-or-text Copilot Studio interview agents in Teams and web chat that write to SharePoint through Power Automate; scoped an onboarding agent (Document Intelligence, Dataverse) for staff from acquired firms.

Stonehage Fleming | Neuchâtel, Switzerland — Mar 2026 – Jun 2026

AI and Automation Analyst (contract), Business Systems

- Built the Local AI SQL Optimiser, a proof of concept ready for production, in which a local model analyses, rewrites and benchmarks SQL queries. Six validation gates and a result-hash equivalence check confirm each rewrite returns the same data, and a transport-level egress allow-list keeps SQL and schema inside the network (FastAPI, Next.js, vLLM, SQLite).
- Designed the firm's AI Innovation Intake platform: a Copilot Studio interview in Teams, Dataverse, Power Automate and branded PRDs generated in LaTeX. Built the FastAPI proof of concept and moved it from Azure OpenAI to a local 14B model; the SQL Optimiser was scoped through a PRD it produced.
- Ran an air-gapped demo for the CTO: four AI tools on one RTX 5080 laptop with networking switched off: OCR of handwritten forms with a 5B open-weight vision model (Chandra-OCR-2), a brief-to-branded-PDF document generator on a local LLM, the intake portal and the SQL Optimiser.
- Wrote a self-hosting research pack of nine papers (use-case matrix, open-model catalogue, fine-tuning, model serving, throughput, platform strategy, infrastructure, OCR cost) behind a GPU-server purchase request.

INDEPENDENT PROJECTS

Theioma | self-hosted multi-agent engineering platform — Mar 2026 – Present

- Built a command centre where one person directs teams of Claude Code and Codex agents by text or voice: an orchestrator plans with the user, project managers staff workers in their own terminals, review and security agents check every finished task, and anything irreversible waits for a passkey-gated approval.
- Built the J-PEX hardware pipeline, which takes a written brief through eight engineering stages (research and parts, CAD, circuits, PCB, firmware, FEA and CFD simulation, documentation, fabrication) to a build manual, using 99 agent tools. Took a drone and a robot-arm brief to build manuals, with OpenFOAM CFD in a network-isolated Docker sandbox.

- Built on Next.js, TypeScript and PostgreSQL with pgvector (219 Prisma models), 24 services, MCP servers with 100+ tools, on-device speech (Whisper, Kokoro) and a mutual-TLS channel to other machines; 6,700+ commits.

Tez | open-source local decision model, github.com/Jibalmi/Tez Sep 2026 – Present

- Built typed decisions with calibrated confidence from one forward pass of a local LLM, an open alternative to TypeSafe's hosted Jev; released it as a Python package, CLI and HTTP server with conformal abstention.
- Measured 0.943 balanced accuracy for Gemma 4 12B on the SemIf benchmark (reference 0.813). A linear probe I trained on the benchmark's labels, on a frozen 4B model's hidden state, reached 0.794 on typed-decisions, ahead of a fine-tuned encoder (0.766) and Jev's published 0.727; 0.885 macro accuracy across 11 languages on MASSIVE (vs 0.524). Replayed voice commands reach the first action 45.5 ms (median) after the decisive word.

Local-first work hub | personal project, built on my own time Jun 2026 – Present

- Built an app for projects, meetings and one-to-ones (Next.js, Prisma, SQLite, Tauri), with an on-device voice assistant (Whisper, local tool-calling model via Ollama, Kokoro), meeting transcription with speaker diarization, and generated minutes and action items. Data lives in a git-versioned vault as an audit trail.

SKILLS

AI and LLMs: vLLM, llama.cpp, Ollama; RAG with pgvector; LoRA/QLoRA fine-tuning; calibration and conformal prediction; OCR vision models; speech (faster-whisper, Kokoro); MCP and multi-agent orchestration

Microsoft and Azure: Azure AI Foundry, Azure OpenAI, Document Intelligence, Azure Policy, Copilot Studio, Power Automate, Dataverse, SharePoint, Microsoft Graph; Laserfiche Workflow

Software: Python (FastAPI, PyTorch, PyMuPDF), TypeScript (Next.js, React, Prisma), C#, SQL, PostgreSQL, SQLite, Docker, Git, LaTeX

Engineering: MATLAB, SolidWorks, Fusion 360, CAD/CAM, FEA, CFD (OpenFOAM)

Languages: English, Portuguese, Spanish (all fluent)

EDUCATION

University of Michigan, College of Engineering | Ann Arbor, MI May 2026
Master's in Space Engineering, GPA 4.01

- **Master's thesis, sponsored by Northrop Grumman:** designed a multi-target active debris removal mission that lets J2 nodal precession align orbital planes in place of costly plane-change burns. Re-evaluated it independently in 2026 on a frozen catalogue snapshot: the median tour reaches 29 rocket bodies in 7 years on 2 km/s, against 9 with impulsive plane changes, and the tour heuristic is within 1.9% of an exact optimum.
- **PERCI, team mission concept (14 students, NASA JPL mentors):** designed a penetrator secondary payload for NASA's Uranus Orbiter and Probe, aimed at Umbriel. My part was science requirements, impact and thermal analysis, trade studies, Ansys and CAD. Re-audited it (61 findings, 4 critical): a central hit needs 330, not 450 m/s.
- **CubeSat attitude-control testbed (SPACE 584, 18 students):** built a 0.95 kg one-axis ADCS that stops and reverses its spin with a reaction wheel. I did the CAD, sensor testing and control logic. Re-audited it (29 findings): showed the filter was over-confident (NEES 187) and raised simulated mission success from 1% to 100% on a v2 design.
- **Nuclear electric propulsion study (SPC 536, two authors, lead writer):** compared ion, Hall and arcjet thrusters on solar, RTG and 100 kWe fission power. Re-audited it (44 issues, 7 critical) and rebuilt it with optimal control.

Boston University, College of Engineering | Boston, MA May 2024
B.S. in Mechanical Engineering (Aerospace), GPA 3.78/4.00, cum laude | Dean's List (6 semesters), Tau Beta Pi

- **Finalist, NASA Gateways to Blue Skies Competition** (2023–24): an aviation-system concept to sample atmospheric conditions around volcanic eruptions and improve ash-cloud forecasts.
- **Rocket Propulsion Club,** Engine Development Team (2021–22): designed an ablative combustion chamber.
- **ME360 builds:** built a three-axis PID camera gimbal (sole designer, CAD to firmware) and a portable three-axis CNC mill; redesigned both in 2026 (brushless gimbal, closed-loop mill).

EARLIER EXPERIENCE

SILab, Boston University | Lab Supervisor Feb 2021 – May 2024
Trained and supervised more than 100 engineering students on workshop safety and machine use.

Galp | Intern, Lisbon Jun 2022
Assessed AI cameras for supervising refinery workers and machines; researched ML-based predictive maintenance.

Hovione FarmaCiência | Lisbon Apr – Aug 2020
Designed a database tracking 40+ tons a week of COVID-19 sanitising gel shipped to public and private charities.

AWARDS, ACTIVITIES AND INTERESTS

Awards: Athlete of the Year (4 times) and 28+ sports awards; Deputy Head Boy, St. Julian's School

Volunteering: Refood, Lisbon (2016–19); Rustic Pathways, Dominican Republic (2017), teaching in rural schools

Interests: rocketry, cosmology, motorsport, European football, cooking, philosophy, the outdoors